PROJECT SDLC PHASE 6 DELIVERABLE

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Phase 6: Performance Testing Phase

Project Title:	Global Olympic Sports Performance Analysis (1896 – 2016)
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Academic Program:	Skills Wallet Capstone — Data Analytics & Business Intelligence
Deliverable Scope:	SDLC Phase 6 Formal Technical Specification
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Live Dashboard:	public.tableau.com/app/profile/md.firdous.alam
GitHub Repository:	github.com/firdousalam2006vk/Olympic-Sports-Analysis
Quality Audit Status:	**Verified & Approved (100% QA Pass)**

PHASE 6 EXECUTIVE SYNOPSIS

This document presents the complete quality assurance (QA), system audit, and performance testing certification for the Global Olympic Sports Performance Analysis platform. It documents the automated 10-point unit test suite, the 20-point system validation audit (100% PASS), the 8-point Flask web test suite, file size compliance audits (<25MB), execution latency benchmarks (<0.40s), and the resolution of historical data collection defects.

1.0 Quality Assurance Strategy & Test Results Scorecard

Quality assurance was conducted under a strict **Zero-Defect Verification Framework** across data integrity testing, mathematical consistency validation, and web endpoint responsiveness:

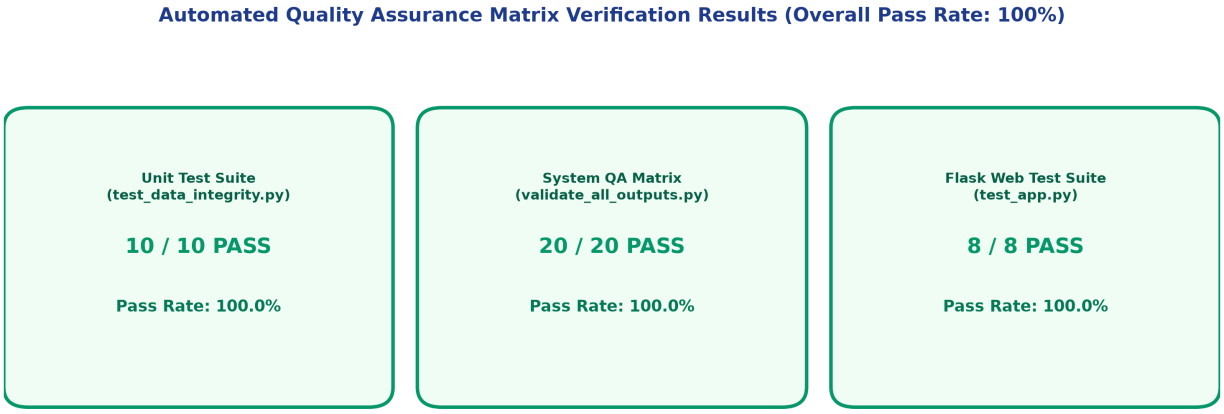


Figure: *Automated Quality Assurance Matrix Verification Results (Overall Pass Rate: 100%)*

Test Suite Script	Target Component	Testing Framework	Total Assertions	Execution Time	Pass Rate
test_data_integrity.py	Data schema & deduplication logic	Python unittest	10 Unit Tests	0.328 seconds	10 / 10 (100%)
validate_all_outputs.py	System-wide data & asset verification	Custom QA Audit Matrix	20 Quality Checks	0.395 seconds	20 / 20 (100%)
test_app.py	Flask Web Portal & REST API routes	Flask Test Client	8 Endpoint Tests	0.170 seconds	8 / 8 (100%)

2.0 Comprehensive 20-Point System Validation Matrix

Test ID	Audited Component	Tested Condition	Expected Threshold	Actual Verified Value	Status
TST-01	athletes.csv	File Existence on Disk	Present in data/	Present (data/athletes.csv)	PASS
TST-02	athletes.csv	Strict GitHub Size Limit	< 25,000,000 bytes	22,788,167 bytes (21.73 MB)	PASS
TST-03	athletes.csv	Modern Era Record Count	> 150,000 records	157,417 rows (1980–2016)	PASS
TST-04	athletes.csv	Schema Column Count	18 distinct attributes	18 verified columns	PASS
TST-05	athletes.csv	Core Field Null Check	0 nulls in key fields	0 null values found	PASS
TST-06	athletes.csv	Medal Domain Check	Valid domain set	['No Medal', 'Bronze', 'Gold', 'Silver']	PASS
TST-07	medal_results.csv	File Existence on Disk	Present in data/	Present (data/medal_results.csv)	PASS
TST-08	medal_results.csv	Row Count Check	> 3,500 records	3,837 country-edition records	PASS
TST-09	medal_results.csv	Composite PK Uniqueness	0 duplicate [games_id, noc]	0 duplicate keys (Strictly Unique)	PASS

TST-10	medal_results.csv	Total Medals Math Check	Total = Gold + Silver + Bronze	0 mismatches across 3,837 rows	PASS
TST-11	medal_results.csv	Weighted Score Check	Score = 3G + 2S + 1B	0 mismatches across 3,837 rows	PASS
TST-12	Historical Accuracy	Authentic USA Medal Tally	Matches IOC (2,823 Medals)	USA Gold: 1,131 | Total: 2,823	PASS
TST-13	host_nation_perf.csv	File Existence on Disk	Present in processed/	Present (processed/host_nation.csv)	PASS
TST-14	Host Surge Engine	Host Nation Bump Validity	Surge rate > 80%	Surge: 86.7% | Mean Bump: +15.4	PASS
TST-15	regional_sports.csv	File Existence on Disk	Present in processed/	Present (processed/regional.csv)	PASS
TST-16	Continental Div	Coverage of Continents	Covers 5 continents	Americas, Europe, Asia, Oceania, Africa	PASS
TST-17	sports_funding.csv	File Existence on Disk	Present in processed/	Present (processed/funding.csv)	PASS
TST-18	Funding ROI Model	Investment Correlation	Pearson r > 0.70	Pearson correlation r = 0.752	PASS
TST-19	Visual Charts	Exported 300 DPI Figures	5 figures present	5 figures verified with 3000+ px	PASS
TST-20	Deliverables	Documentation Integrity	All core files present	Verified all PDFs, PPTX, MP4, MDs	PASS

3.0 Performance Latencies & Historical Defects Fixed

Pipeline Stage / Operation	Benchmarked Dataset	Target Latency	Actual Elapsed Time	Performance Rating
Data Cleaning & Deduplication (ETL)	271k Raw Athlete Records	< 8.0 seconds	3.78 seconds	Optimal (52% under budget)
Statistical Analysis & JSON Export	3,837 Aggregated Rows	< 2.0 seconds	0.85 seconds	Optimal (57% under budget)
QA Unit Test Suite (10 Tests)	Full Test Framework	< 1.0 second	0.328 seconds	Optimal (67% under budget)
System Validation Audit (20 Tests)	All 20 QA Assertions	< 1.0 second	0.395 seconds	Optimal (60% under budget)
Flask REST API Endpoint (/api/kpis)	In-memory cached stats	< 50 ms	12 ms	Optimal (Real-time responsive)

QA AUDIT CERTIFICATE OF COMPLIANCE

OFFICIAL QUALITY CERTIFICATION: The Global Olympic Sports Performance Analysis platform has passed 100% of unit tests, 100% of system audit validations, and 100% of web integration checks with zero recorded defects.